

Gayathri Akkinapalli

(413) 466-1166 | gayathri.akkinapalli2000@gmail.com | [linkedin.com/in/gayathri-akkinapalli](https://www.linkedin.com/in/gayathri-akkinapalli) | github.com/akkina10gayu | [Google Scholar](https://scholar.google.com/citations?user=akkinapalli)

Summary

Python and AI/ML Engineer with 4+ years of experience building production ML systems, search and retrieval pipelines, recommendation models, and LLM applications. Experienced in developing agentic AI and RAG systems, LLM evaluation and alignment, NLP, prompt engineering, tool-integrated workflows, and scalable ML infrastructure. Strong background in taking ML solutions from experimentation and evaluation through production deployment and optimization.

Education

University of Massachusetts Amherst (UMASS) | MS in Computer Science | CGPA: 4.0/4.0 Sep 2023 - May 2025
Relevant Coursework: Advanced Natural Language Processing (NLP), Reinforcement Learning (RL), Intelligent Visual Computing, Systems for Data Science, Responsible AI, Algorithms for Data Science, Statistics for Applied Math, Business Intelligence and Analytics, Deep Learning, Independent Study

Indian Institute of Information Technology (IIIT) | B.Tech in Computer Engineering | CGPA: 9.07/10.0 Aug 2017 - May 2021
Relevant Coursework: Discrete Structures and Algorithms, Database Management Systems, Artificial Intelligence, Foundations of Machine Learning, Statistical Learning

Skills

Programming Languages: Python, R, SQL | Familiar: Java, React JS, C, C++

Tools/Libraries: LangChain, MCP, Unsloth, Wandb, Keras, PyTorch, TensorFlow, PySpark, SKLearn, NumPy, vLLM, OpenAI, Copilot

Software/Frameworks: Docker, Kubernetes, Kafka, REST API, Splunk, MongoDB, Tableau, CI/CD, Snowflake, FAISS, AWS, GCP

Experience

Oracle Remote, US

Machine Learning Engineer (Contract) Aug 2025 - Present

- Built a hybrid search pipeline combining dense retrieval via Oracle AI Vector Search and BM25 sparse search, fused via Reciprocal Rank Fusion, improving Recall@20 by 26% over the keyword-only baseline.
- Fine-tuned MonoBERT re-ranker on 200K query-click pairs to reorder top-50 fused candidates, improving MRR by 18%.
- Implemented query normalization for noisy and verbose inputs, reducing zero-result rate from 7.2% to 3.1% across 30K daily queries.
- Evaluated retrieval quality using Recall, NDCG on a 3K human-ranked judgments, with weekly regression benchmarks on MLflow.
- Deployed with FastAPI, Kubernetes, and Redis caching, optimizing first-stage retrieval for p99 latency under 80ms.
- Developed an agentic AI workflow using OpenAI function calling and structured orchestration to detect, create, and route customer incident tickets, reducing manual triage time by 35% and achieving 92% routing accuracy across 100+ daily tickets.

Cisco Amherst, MA, US

Graduate Researcher Jan 2025 - Jul 2025

- Developed PerFine, a training-free critique-refine Agentic RAG framework for enhancing personalization in long-text using LangChain.
- Evaluated with LLM-as-a-Judge (G-Eval), improving personalization by 13% and Meteor by 10% over RAG-based baselines.
- Enhanced outputs with profile-grounded critic feedback retrieved via Pinecone, FAISS, MCP refining style, content relevance.

UMass IESL Lab Amherst, MA, US

Research Assistant Aug 2024 - Dec 2024

- Implemented an autoregressive model that performs lookahead by decoding in superposition with 2 forward passes using cross attention.
- Applied approach to machine translation, improving BLEU score and generation quality of MT5 model by approximately 15%.

Carelon Global Solutions Hyderabad, TG, India

Data Scientist Jun 2021 - Jul 2023

- Built Recommendation Systems using NER, SpaCy & ML models like XGBoost in PySpark improving NDCG@5 by 75%.
- Designed and deployed end-to-end ML pipelines with REST APIs, Hive, MongoDB, Redis, reducing careplan creation time by 2 hours.
- Performed Aspect-Based Sentiment Analysis on call transcripts with RoBERTa and SpaCy models achieving 85% accuracy & 0.81 F1-score.
- Engineered Conversational AI bot using DistilBERT for intent classification and BERT QA with RAG over S3, deployed on AWS SageMaker, resolving 65% of user queries across 1000+ daily interactions.
- Integrated web-scraped healthcare articles into Elasticsearch with ranking optimization, reducing content retrieval time by 120ms.
- Orchestrated model deployment with Kubernetes, Kafka, CI/CD pipelines, reducing deployment time by ~70%, improving scalability.

- Created Splunk dashboards and conducted A/B testing for feedback-driven model improvements, achieving 60% performance gain.
- Engineered AWS Lambda functions to extract conversational AI metrics from S3 into DynamoDB and AWS Glue ETL pipelines to transform and migrate conversation data.

SensorDrops Networks (STEP at IIT Kharagpur)

Remote, India

AI Engineer Intern

Aug 2020 - Sep 2020

- Modeled real-time Social Distance Monitoring system leveraging YOLOv3 with live feed, bounding boxes, 90% detection accuracy.
- Deployed application on AWS EC2 and Docker, enabling real-time analytics with ~200ms latency via socket-based data transfer.

Centre for Development of Advanced Computing (C-DAC)

Pune, MH, India

AI Engineer Intern

May 2020 - Aug 2020

- Created a prototype of customized deep CNN model to identify COVID-infected chest X-rays with accuracy of around 92%.
- Trained model on High Performance Computing (HPC) for three chest X-ray classes, obtaining validation F1-score of 0.9.

Projects

EduNotes: AI Study Assistant | [Github](#)

- Designed a multi-agent RAG study assistant, centralizing study materials into a queryable KB, using intelligent agents for search, content processing and adaptive summarization with Llama-70B/8B/Scout, Gemini-2.5 Flash, LangChain, and ChromaDB.
- Implemented flashcards, quizzes, notes, summaries, and conversational AI chat with 7 learning modes via FastAPI & Streamlit, supporting 6+ input types (PDF, URL, text, topic, web search) with Research Mode for academic references and PDF figure, table analysis.

Aligning LLMs towards safety and helpfulness | UMass | [Github](#)

- Aligned LLaMa-2 toward safety using LoRA, QLoRA on PKU-SafeRLHF benchmark with SFT, RAFT, RLHF, DPO in TRL.
- Scored 93% safe on DPO (40% SFT) with Llama-Guard on I-CoNa. Employed LLM-as-a-judge to evaluate safety and helpfulness.

Human Motion Synthesis with Latent-space GANs | UMass | [Github](#)

- Generated text-to-motion sequences in latent space utilizing GANS, VAE, CLIP on HumanML3D with Distributed training in lightning.
- Secured FID of 0.48 for GAN in latent space with 91% FLOPs reduction compared to Latent Diffusion Model on HumanML.

Optimization in Reinforcement Learning | UMass | [Github](#)

- Programmed Reinforce with baseline, Actor-Critic, PPO & Semi Gradient n-step SARSA for Acrobat and Cartpole environments.
- Attained stabilized mean rewards of 470 (max = 500), -100 (max = 0) on Cartpole & Acrobat respectively with Reinforce, Actor-Critic.

Real time Stock Analysis | UMass | [Github](#)

- Devised a Kafka-PySpark pipeline for stock news analysis using LLaMA, achieving 84% GPT-4 alignment and 40% faster processing.
- Integrated a hybrid RAG retrieval system (dense + keyword search) for financial information, improving accuracy by an additional 5%.

Publications & Preprints

- LS-GAN: Human Motion Synthesis with Latent-space GANs *IEEE WACVW '25*
- PerFine: Iterative Critique-Refine Framework for Enhancing LLM Personalization *AACL-IJCNLP 2026*
- Safe to Serve: Aligning Instruction-Tuned Models for Safety and Helpfulness *arXiv:2412.00074*
- Automated Model Selection for Tabular Data *arXiv:2401.00961*

Achievements & Awards

- **Guinness World Record awardee** for largest Kuchipudi dance with 2,850 participants at International Kuchipudi Dance Convention.
- **Honored with Batch Topper Award** at IIIT Kurnool for ranking 1st among all students across departments.
- **Received the Go Above Impact Award** twice at Carelon Global Solutions for exceptional contributions to the team.